MEDIUM

INFORMATION AND IDEAS

Command of Evidence

07

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Characteristics of Five Recently Discovered Gas Exoplanets

Exoplanet designation	Mass (Jupiters)	Radius (Jupiters)	Orbital period (days)	Distance from the Sun (parsecs)
TOI-640 b	0.88	1.771	5.003	340
TOI-1601 b	0.99	1.239	5.331	336
TOI-628 b	6.33	1.060	3.409	178
TOI-1478 b	0.85	1.060	10.180	153
TOI-1333 b	2.37	1.396	4.720	200

"Hot Jupiters" are gas planets that have a mass of at least 0.25 Jupiters (meaning that their mass is at least 25% of that of Jupiter) and an orbital period of less than 10 days (meaning that they complete one orbit around their star in less than 10 days), while "warm Jupiters" are gas planets that meet the same mass criterion but have orbital periods of more than 10 days. In 2021, Michigan State University astronomer Joseph Rodriguez and colleagues announced the discovery of five new gas exoplanets and asserted that four are hot Jupiters and one is a warm Jupiter.

Which choice best describes data from the table that support Rodriguez and colleagues' assertion?

A. None of the planets have an orbital period of more than 10 days, and TOI-628 b has a mass of 6.33 Jupiters.

B. TOI-1478 b has an orbital period of 153 days, and the masses of all the planets range from 0.85 to 6.33 Jupiters.

C. All the planets have a radius between 1.060 and 1.771 Jupiters, and only TOI-1333 b has an orbital period of more than 10 days.

D. Each of the planets has a mass greater than 0.25 Jupiters, and all except for TOI-1478 b have an orbital period of less than 10 days.

Q2

INFORMATION AND IDEAS

Many music streaming services also function as social media platforms: by giving users the option to follow friends and share curated playlists or information about their listening history, these platforms allow people to convey their music preferences and listening activities directly to their social networks. In a 2023 study, researcher Michael James Walsh interviewed frequent users of a popular music streaming platform to investigate how its social media dimensions shape their listening habits. Walsh found that these dimensions tended to make study participants feel more mindful of how their listening activities may be perceived, which in turn influenced how they managed those activities.

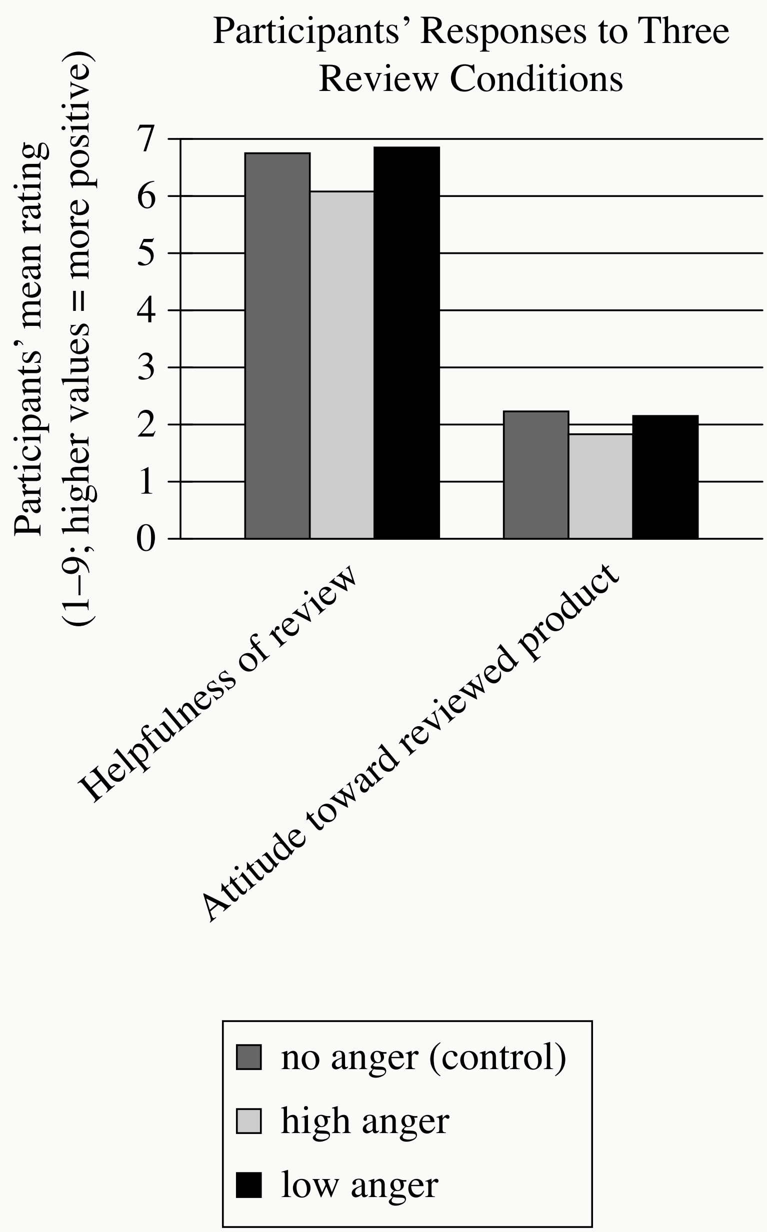
Which quotation from a music streaming platform user would best illustrate Walsh's finding?

A. "When I create playlists or choose songs to listen to, I often pick songs that I think the friends who are following me might be interested in listening to, rather than what I might actually prefer listening to in that moment."

B. "Because the predictions the platform makes about what new songs I might like depend on my past listening habits, at times its recommendations can feel overly repetitive and unvarying."

C. "The social aspects of music streaming make it really easy to create a sense of a shared listening experience with my friends, even if we aren't in the same room."

D. "Listening to music through a streaming platform feels a lot more passive, since I don't need to make any deliberate choices about what music I'm going to consume, like I do when buying digital downloads or records and CDs."



- For each data category, the following bars are shown:
  - no anger (control)
  - high anger
  - low anger
- The data for the 2 categories are as follows:
  - Helpfulness of review:

        - no anger (control): 6.75
        - high anger: 6.08
        - low anger: 6.85
    - Attitude toward reviewed product:
        - no anger (control): 2.23
        - high anger: 1.83
        - low anger: 2.15

To understand how expressions of anger in reviews of products affect readers of those reviews, business scholar Dezhi Yin and colleagues measured study participants' responses to three versions of the same negative review—a control review expressing no anger, a review expressing a high degree of anger, and a review expressing a low degree of anger. Reviewing the data, a student concludes that the mere presence of anger in a review may not negatively affect readers' perceptions of the review, but a high degree of anger in a review does worsen readers' perceptions of the review.

Which choice best describes data from the graph that support the students' conclusion?

A. On average, participants' ratings of the helpfulness of the review were substantially higher than were participants' ratings of the reviewed product regardless of which type of review participants had seen.

B. Compared with participants who saw the control review, participants who saw the low-anger review rated the review as slightly more helpful, whereas participants who saw the high-anger review rated the review as less helpful.

C. Participants who saw the low-anger review rated the review as slightly more helpful than participants who saw the control review did, but participants' attitude toward the reviewed product was slightly worse when participants saw the low-anger review than when they saw the no-anger review.

D. Compared with participants who saw the low-anger review, participants who saw the high-anger review rated the review as less helpful and had a less positive attitude toward the reviewed product.

Q4

INFORMATION AND IDEAS

"The Yellow Wallpaper" is an 1892 short story by Charlotte Perkins Gilman. In the story, the narrator expresses mixed feelings about her surroundings: ______blank

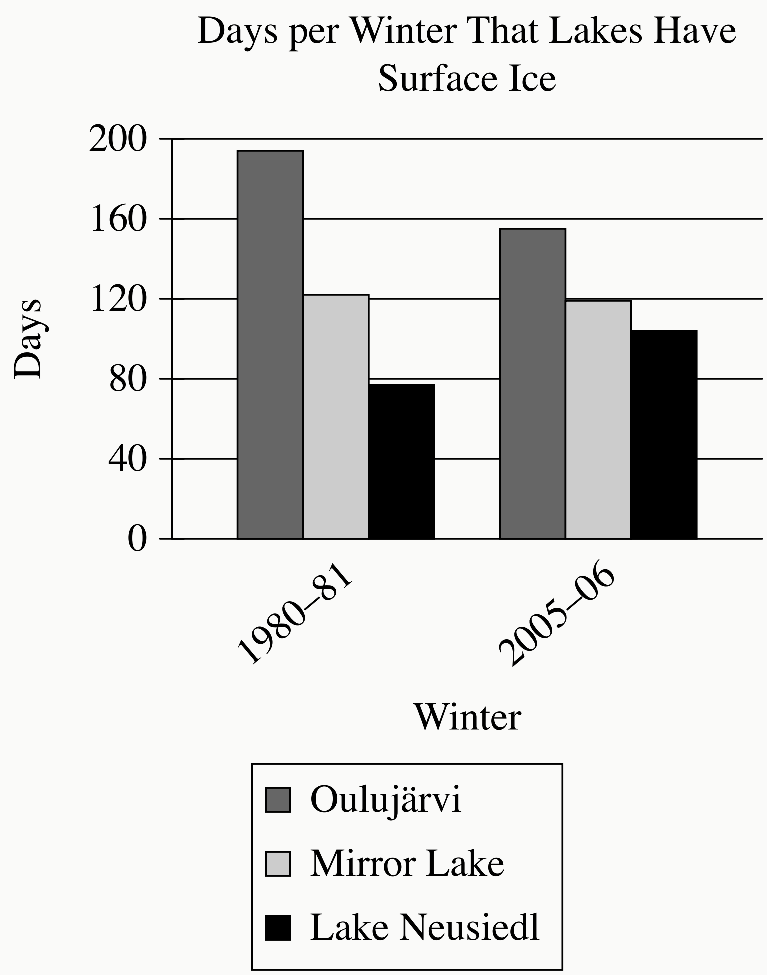
Which quotation from "The Yellow Wallpaper" most effectively illustrates the claim?

A. "This wallpaper has a kind of sub-pattern in a different shade, a particularly irritating one, for you can only see it in certain lights, and not clearly then."

B. "By moonlight—the moon shines in all night when there is a moon—I wouldn't know it was the same paper."

C. "I'm really getting quite fond of the big room, all but that horrid [wall]paper."

D. "The color is repellant, almost revolting; a smouldering, unclean yellow, strangely faded by the slow-turning sunlight."



- For each data category, the following bars are shown:
  - Oulujärvi
  - Mirror Lake
  - Lake Neusiedl
- The data for the 2 categories are as follows:
  - 1980–81:
    - Oulujärvi: 194 days
    - Mirror Lake: 122 days
    - Lake Neusiedl: 77 days
  - 2005–06:
    - Oulujärvi: 155 days

- Mirror Lake: 119 days
- Lake Neusiedl: 104 days

It is common for freshwater lakes near or above a latitude of 45° north of the equator, like Lake Mjøsa in Norway, to accumulate surface ice in winter. The amount and duration of ice depends on many factors, including local weather conditions as well as the lake's depth, volume, and surface area, but a climate researcher claims that some lakes in these latitudes have seen a decline in the duration of ice between the early 1980s and the mid-2000s. She cites as a typical example ______blank

Which choice most effectively uses data from the graph to complete the researcher's example?

A. both Lake Neusiedl and Oulujärvi, which had fewer than 195 days of ice in the winter of 1980–81.

B. Lake Neusiedl, which had more days of ice in the winter of 2005–06 than it did in the winter of 1980–81.

C. Oulujärvi, which had fewer days of ice in the winter of 2005–06 than it did in the winter of 1980–81.

D. both Lake Neusiedl and Oulujärvi, which had more than 105 days of ice in the winter of 2005–06.

Q6 INFORMATION AND IDEAS

Puerto Rico is an island in the Caribbean Sea. Indigenous people there started raising guinea pigs about 1,700 years ago. Guinea pigs had originally been domesticated much earlier in both Colombia and Peru. So were guinea pigs brought to Puerto Rico from Colombia or from Peru? Ancient Caribbean trade routes connected Puerto Rico with Colombia but not with Peru. <u>Therefore, guinea pigs in Puerto Rico probably came from Colombia and descended from Colombian guinea pigs.</u>

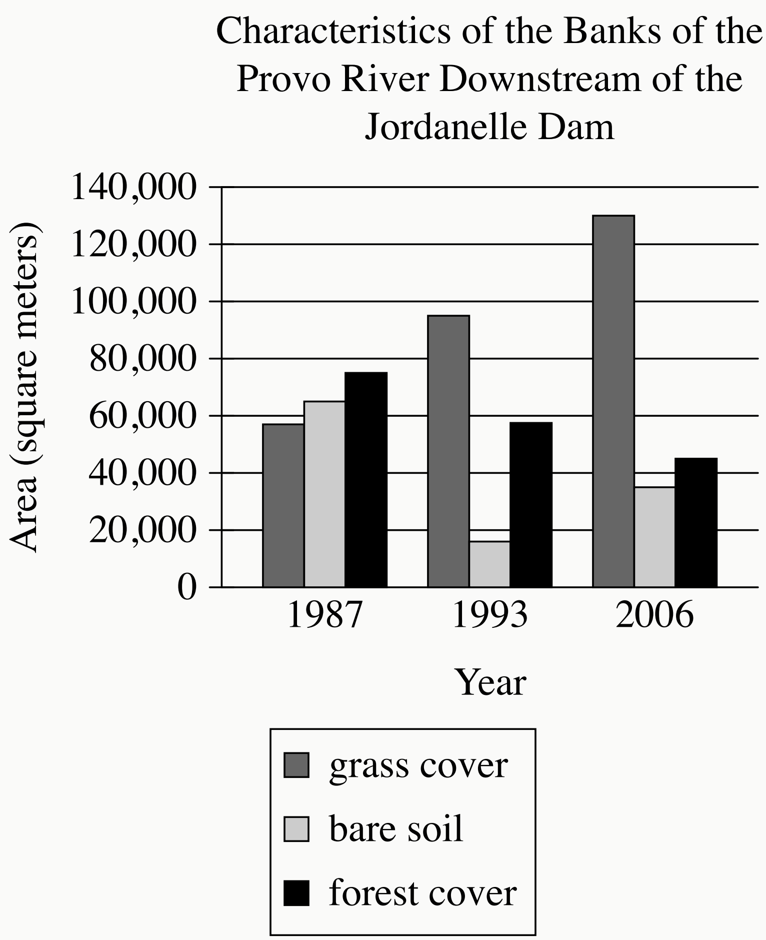
Which finding, if true, would most directly weaken the underlined claim?

A. Ancient guinea pigs in Puerto Rico were genetically less similar to ancient guinea pigs in Colombia than to ancient guinea pigs in Peru.

B. Guinea pigs are common in ancient Puerto Rican art, especially in pottery.

C. Modern breeds of guinea pigs don’t look like images of guinea pigs in ancient art from Puerto Rico, Colombia, and Peru.

D. The guinea pig population of ancient Colombia was much larger than the guinea pig population of ancient Peru.



- For each data category, the following bars are shown:
  - grass cover
  - bare soil
  - forest cover
- All values are approximate.
- The data for the 3 categories are as follows:
  - 1987:
    - grass cover: 57,000 square meters
    - bare soil: 65,000 square meters
    - forest cover: 75,000 square meters
  - 1993:
    - grass cover: 95,000 square meters

        - bare soil: 16,000 square meters
        - forest cover: 57,500 square meters
    - 2006:
        - grass cover: 130,000 square meters
        - bare soil: 35,000 square meters
        - forest cover: 45,000 square meters

The Jordanelle Dam was built on the Provo River in Utah in 1992. Earth scientist Adriana E. Martinez and colleagues tracked changes to the environment on the banks of the river downstream of the dam, including how much grass and forest cover were present. They concluded that the dam changed the flow of the river in ways that benefited grass plants but didn't benefit trees.

Which choice best describes data from the graph that support Martinez and colleagues' conclusion?

A. The lowest amount of grass cover was approximately 58,000 square meters, and the highest amount of forest cover was approximately 75,000 square meters.

B. There was more grass cover than forest cover in 1987, and this difference increased dramatically in 1993 and again in 2006.

C. There was less grass cover than bare soil in 1987 but more grass cover than bare soil in 1993 and 2006, whereas there was more forest cover than bare soil in all three years.

D. Grass cover increased from 1987 to 1993 and from 1993 to 2006, whereas forest cover decreased in those periods.

Although most songbirds build open, cupped nests, some species build domed nests with roofs that provide much more protection. Many ecologists have assumed that domed nests would provide protection from weather conditions and thus would allow species that build them to have larger geographic ranges than species that build open nests do. To evaluate this assumption, a research team led by evolutionary biologist Iliana Medina analyzed data for over 3,000 species of songbirds.

Which finding from Medina and her colleagues' study, if true, would most directly challenge the assumption in the underlined sentence?

A. Species that build open nests tend to have higher extinction rates than species that build domed nests.

B. Species that build open nests tend to be smaller in size than species that build domed nests.

C. Species that build open nests tend to use fewer materials to build their nests than species that build domed nests do.

D. Species that build open nests tend to have larger ranges than species that build domed nests.

Q9 INFORMATION AND IDEAS

Participants' Evaluation of the Likelihood That Robots Can Work Effectively in Different Occupations

Occupation	Somewhat or very unlikely (%)	Neutral (%)	Somewhat or very likely (%)
television news anchor	24	9	67
teacher	37	16	47
firefighter	62	9	30
surgeon	74	9	16
tour guide	10	8	82

Rows in table may not add up to 100 due to rounding.

Georgia Tech roboticists De'Aira Bryant and Ayanna Howard, along with ethicist Jason Borenstein, were interested in people's perceptions of robots' competence. They recruited participants and asked them how likely they think it is that a robot could do the work required in various occupations. Participants' evaluations varied widely depending on which occupation was being considered; for example, ______blank

Which choice most effectively uses data from the table to complete the example?

A. 47% of participants believe that it is somewhat or very likely that a robot could work effectively as a teacher, but 37% of respondents believe that it is somewhat or very unlikely that a robot could do so.

B. 9% of participants were neutral about whether a robot could work effectively as a television news anchor, which is the same percent of participants who were neutral when asked about a robot working as a surgeon.

C. 82% of participants believe that it is somewhat or very likely that a robot could work effectively as a tour guide, but only 16% believe that it is somewhat or very likely that a robot could work as a surgeon.

D. 62% of participants believe that it is somewhat or very unlikely that a robot could work effectively as a firefighter.

Q10

INFORMATION AND IDEAS

Rivers rich in sediment appear yellow, while increases in red algae make rivers appear red. To track things like the sediment or algae content of large US rivers, John R. Gardner and colleagues used satellite data to determine the dominant visible wavelengths of light measured for various segments of these rivers. The researchers classified wavelengths of 495 nanometers (nm) and below as red, wavelengths between 495 and 560 nm as blue, and wavelengths of 560 nm and above as yellow. The researchers concluded that for the Missouri River, segments flowing into lakes tend to carry more sediment than those flowing out of lakes.

Which finding, if true, would most directly support the researchers' conclusion?

A. The segments of the Missouri River that had higher levels of chlorophyll-a, which contributes to the green color of photosynthetic organisms, have dominant wavelengths of light between 490 and 560 nm.

B. In lakes through which segments of the Missouri River pass, the dominant wavelength of light tended to be above 560 nm near the lakes' shores and below 560 nm in the lakes' centers.

C. The majority of the segments of the Missouri River were found to have dominant wavelengths of light significantly higher than 560 nm.

D. Segments of the Missouri River flowing into lakes typically had dominant wavelengths of light above 560 nm, while segments flowing out of lakes typically had dominant wavelengths below 560 nm.